

Personal Project - Project Plan

Cinna's PAH (Protective Afro-Hairstyle Visualizer & Calculator)

Client: Fontys University of Applied Sciences / Personal Software Project

Location: Eindhoven, Netherlands

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Status: Active / Refined

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Version History

Version	Date	Author	Changes	Status
1.0	Feb 16, 2026	Cinna	Initial Project Proposal & PoC continuation from last semester	Done
2.0	April 21, 2026	Cinna	Transition to R3F and sprint planning	Done
2.2	May 8, 2026	Cinna	Refining Architecture & Technical Strategy	Active

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1. Project Assignment

1.1 Context

In the Afro-hair care community, "protective styling" (braids, locs, twists) is a fundamental practice. However, a persistent problem remains: the inability to accurately estimate how many packs of hair extensions are required. Users currently rely on inconsistent social media advice, leading to over-purchasing (wasted money) or under-purchasing (running out of hair mid-process). This project aims to digitize this experience-based knowledge through a 3D web application.

1.2 Project Goal

The goal is to move beyond the Semester 2 Proof of Concept (PoC) into a professional, interactive tool that allows users to:

1. Visualize hairstyles on a 3D head model in real-time.
2. Receive an accurate calculation of required hair packs based on specific density and style parameters.
3. Reduce the stress and financial waste associated with manual estimation.

KPI Table

Area	KPI
Performance	30 FPS minimum on mobile
Accuracy	± 1 hair pack difference from stylist estimate
UX	User can configure hairstyle in under 2 minutes
Stability	No critical rendering crashes during testing
Deployment	Publicly accessible Vercel deployment

1.3 Assignment

The assignment is to develop a functional web application using **React 19** and **React-Three-Fiber (R3F)**. The product must include:

- A responsive UI with parameter sliders (Thickness, Length, Density).
- A procedural 3D hair generation system.
- A validated calculation formula researched last semester using peer feedback and comparative braid chart analysis:

$$(\text{style} + \text{thickness} + \text{density}) \times \text{length} \times 0.95.$$

Variable	Meaning
<i>style</i>	Base complexity coefficient
<i>thickness</i>	Strand size multiplier
<i>density</i>	Scalp coverage multiplier
<i>length</i>	Hair extension consumption factor
<i>0.95</i>	The coefficient (0.95) acts as a normalization factor and may change during validation.

Validation Method

- Compare calculator estimate against:
 - stylist recommendation
 - real-world hairstyle installations
 - braid chart references

Acceptance Threshold

- Estimate deviation should remain within ± 1 pack for common styles.

1.4 Scope

Included in the project:

- Migration from Fontys GitLab to an open-source GitHub repository.
- 3D Head Model ([custombust.glb](#)) and hair segment assets.
- Procedural placement logic using Raycasting and UV Scalp Masking.
- Zustand-based state management.
- Deployment via Vercel.

Excluded from the project:

- A full commercial backend (initial storage will be [localStorage](#)).
- In-app purchasing/E-commerce functionality.
- Physics-based hair simulation (hair will be static/rigid).

Functional Requirements

ID	Requirement
FR-1	User can rotate 3D head model
FR-2	User can change hairstyle density
FR-3	Application recalculates pack estimate in real time
FR-4	User settings persist in localStorage
FR-5	Application works on mobile and desktop

Non-Functional Requirements

ID	Requirement
NFR-1	Initial load under 5 seconds
NFR-2	Maintain ≥ 30 FPS during interaction
NFR-3	Responsive on screens from 360px width
NFR-4	GLTF assets optimized under 5-10 MB or less

1.5 User Personas

I have developed three distinct user personas, based on user interviews conducted during semester 1 at Fontys, that capture the technical, cultural, and professional diversity of the target audience. This includes the specific addition of a male user to address short-style visualization and terminology barriers.

Persona 1: DIY Braider "Budget Strategist"

- **Profile:** Maya, a 22-year-old university student.
- **Context:** Maya braids her own hair to save money. She often finds herself at the beauty supply store guessing how many packs of "X-expression" hair she needs.
- **Pain Point:** She frequently runs out of hair halfway through her head (back-of-the-head panic) or buys two packs too many that sit in her closet for months.
- **Tech Needs:** Mobile-first interface; needs to see how "Small" vs. "Medium" knotless braids affect the pack count.

Persona 2: "Visual Explorer"

- **Profile:** Jordan, a 25-year-old male exploring natural hair growth.
- **Context:** Jordan is transitioning from a fade to short twists or "starter locs." He doesn't know the difference between "two-strand twists" and "finger coils", he just knows what he wants it to look like.
- **Pain Point:** Difficulty communicating with stylists due to lack of terminology. He wants to know if his current hair density is enough for a specific look.
- **Tech Needs:** Visual presets (choosing by image/3D model rather than text); focus on shorter hair lengths and scalp visibility (parting patterns).

Persona 3: Professional Stylist

- **Profile:** Sarah, owner of a specialized Afro-hair boutique in Eindhoven.
- **Context:** Sarah sees 4-6 clients a day. She uses the tool during consultations to provide accurate price quotes based on material usage.
- **Pain Point:** Clients often bring the wrong type or insufficient amount of hair to appointments, causing delays and friction.
- **Tech Needs:** High accuracy; ability to show clients the "fullness" of a style virtually before starting the 6-hour labor process.

Comparative Persona Matrix

Feature	Persona 1: DIY Braider	Persona 2: Male Explorer	Persona 3: Pro Stylist
Primary Goal	Prevent under-purchasing hair.	Visualize short styles & learn terms.	Standardize client consultations.
Main Challenge	Financial waste & mid-style stress.	Terminology gap (Locs vs. Twists).	Inconsistent client preparation.
Key UI Interaction	Pack Calculation Result.	3D Style Gallery/Presets.	Density & Thickness Sliders.
Preferred Device	Mobile (while in-store).	Mobile (for quick reference).	Tablet/Laptop (in-salon).
Tech Literacy	High (TikTok/Social native).	Medium (Visual learner).	High (Business-oriented).
Critical Requirement	Accurate "Style-to-Pack" ratio.	High-fidelity scalp/parting detail.	Formula reliability & Speed.

1.6 Constraints

- **Technology:** Must use React 19 and Node.js. No PostCSS (project preference).
- **Time:** Project must be finalized within the Semester 3 timeline, spending about 8h per week. In my case condensed in sprint 3 to 5 to prioritize high-fidelity rendering logic over early-stage administrative tasks.

1.7 Research Questions

- **Main Question:** How can procedural 3D visualization and mathematical modeling accurately predict hair extension requirements for diverse Afro-hairstyles?
- **Sub-question 2:** How can procedural generation techniques reduce manual placement complexity in 3D hairstyle visualization?
- **Sub-question 3:** How does real-time visualization affect user confidence in estimating extension requirements?
- **Sub-question 4:** What rendering strategies provide the best balance between visual density and real-time performance in browser-based procedural hair systems?

1.8 Final Products

1. **Functional Web App:** Hosted on Vercel.
2. **Project Portfolio:** Documenting the transition from Next.js to React and the Three.js self-study.
3. **Technical Documentation:** Scalp masking logic and Raycasting algorithm overview.
4. **Validation Report:** Results from expert hairstylist interviews.

2. Approach and Planning

2.1 Approach

The project follows an **Agile methodology** with a significant pivot occurring in Week 10.

- **Pivot Justification:** Initial development was delayed due to "planning paralysis" caused by an overly complex architectural end-goal. To maintain momentum and meet Semester 3 requirements, the project was refocused on a **"Utility-First" PoC**.
- **Strategy Shift:** The focus shifted from achieving perfection to **iterative functional delivery**, learning to build and validate core 3D systems even if they require further optimization later. This allowed for a condensed, high-intensity development cycle starting in the second half of the semester.

Research Methods

- **Prototyping:** Testing different Raycasting intervals to balance visual density with performance.
- **Expert Interview:** Structured interviews with Eindhoven-based Afro-hairstylists to validate the calculation formula.
- **Literature/Tutorial Review:** Deep dive into R3F and InstancedMesh documentation (building on previous semester Three.js study).

Project Breakdown

- **Phase 1:** Asset preparation, GLTF optimization, and GitHub environment setup.
- **Phase 2:** Developing the "Parting" algorithm and UV-based spawn logic.
- **Phase 3:** Glassmorphism interface implementation with Tailwind CSS v4.

2.2 Schedule

Note: This schedule accounts for the breaks seen in the calendar (16-02 and 27-04).

Sprint	Deliverables	Effort per week	Start Date	Exit Criteria	Status
S1-S2	Research & Inspiration	4h	Feb 9	Conceptualization of 3D Pipeline	Done
-	<i>Spring Break</i>	<i>0h</i>	<i>Feb 16</i>	<i>No academic activity</i>	-
S3.1	Setup & Migration	8h	Apr 13	App successfully deploys on Vercel	Done
S3.2	3D Core: "Parting"	8h	Apr 20	Raycasting logic spawns mesh instances	In Progress (debug)
-	<i>May Break</i>	<i>8h</i>	<i>Apr 27</i>	<i>Created custom 3D model in Blender for bust and hair</i>	Done
S3.3	Logic Sync	8h	May 4	Zustand state links sliders to 3D	In Progress
S4	Validation & UI Polish	16h	May 11	3+ Stylist feedback sessions	Pending
S5	Documentation & Demo	16h	June 8	Portfolio and technical doc finalized	Pending
Final	Handover / Endtalks	4h	June 22	Project presented	Pending

Developer's Note on Timeline:

The project timeline reflects a strategic delay during Sprints 1 and 2. This was a result of high conceptual complexity leading to 'inspiration block.' By Sprint 3, the project was restructured to prioritize a 'Build-to-Learn' philosophy. This allowed me to overcome the pressure of meeting an insanely complicated end-goal by breaking the technical requirements into manageable 8-hour weekly sprints, focusing on functional procedural generation over perfection.

3. Technical Architecture Diagram

3.1 System Architecture

The system architecture defines the relationship between the client-side UI, the state management layer, and the 3D rendering engine.

Core Tech Stack

- **Frontend Framework:** React 19 (UI Logic & Component Lifecycle)
- **State Management:** Zustand (Global store for hairstyle parameters)
- **3D Engine:** React-Three-Fiber & Three.js (Procedural mesh generation)
- **Build Tool:** [Node.js](#) (Fast HMR and optimized bundling)
- **Deployment:** Vercel (CI/CD and public hosting)

Planned Architectural Layers

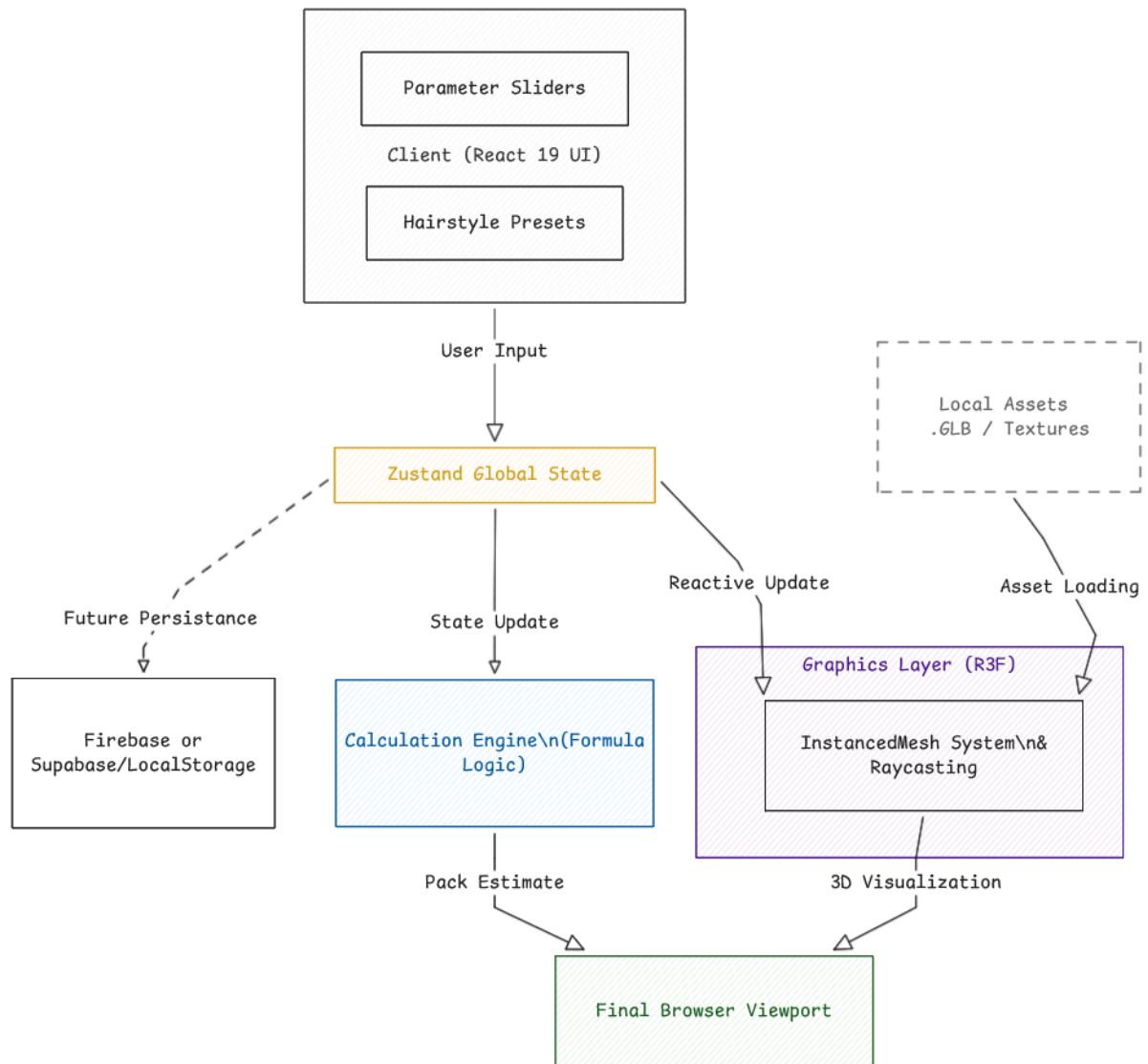
- **Presentation Layer:** React components handling the Glassmorphism UI and slider inputs.
- **State Layer:** A centralized Zustand store that acts as the "Single Source of Truth" for hair density, length, and style coefficients.
- **Logic Layer:** The Calculation Engine that processes the mathematical formula and the Raycasting algorithm for mesh placement.
- **Graphics Layer:** The R3F Canvas where the optimized GLB bust and instanced hair segments are rendered.

Purpose

The architecture is designed to be decoupled and performant:

- **Decoupling:** By separating the Calculation Engine from the Rendering Layer, the mathematical estimates can update instantly even if the 3D scene is still processing complex mesh updates.
- **Performance:** Leveraging a "push-based" state update (Zustand) ensures that only the necessary 3D components re-render when a slider is moved, maintaining the target 30+ FPS.
- **Scalability:** The structure allows for an easy transition from localStorage to a backend database (Supabase) in future iterations without refactoring the core rendering logic.

System Architecture Diagram (Sprint 3)



3.2 UML Component Diagram

The UML Component Diagram will visualize the relationship between the main systems of the application.

Planned Components

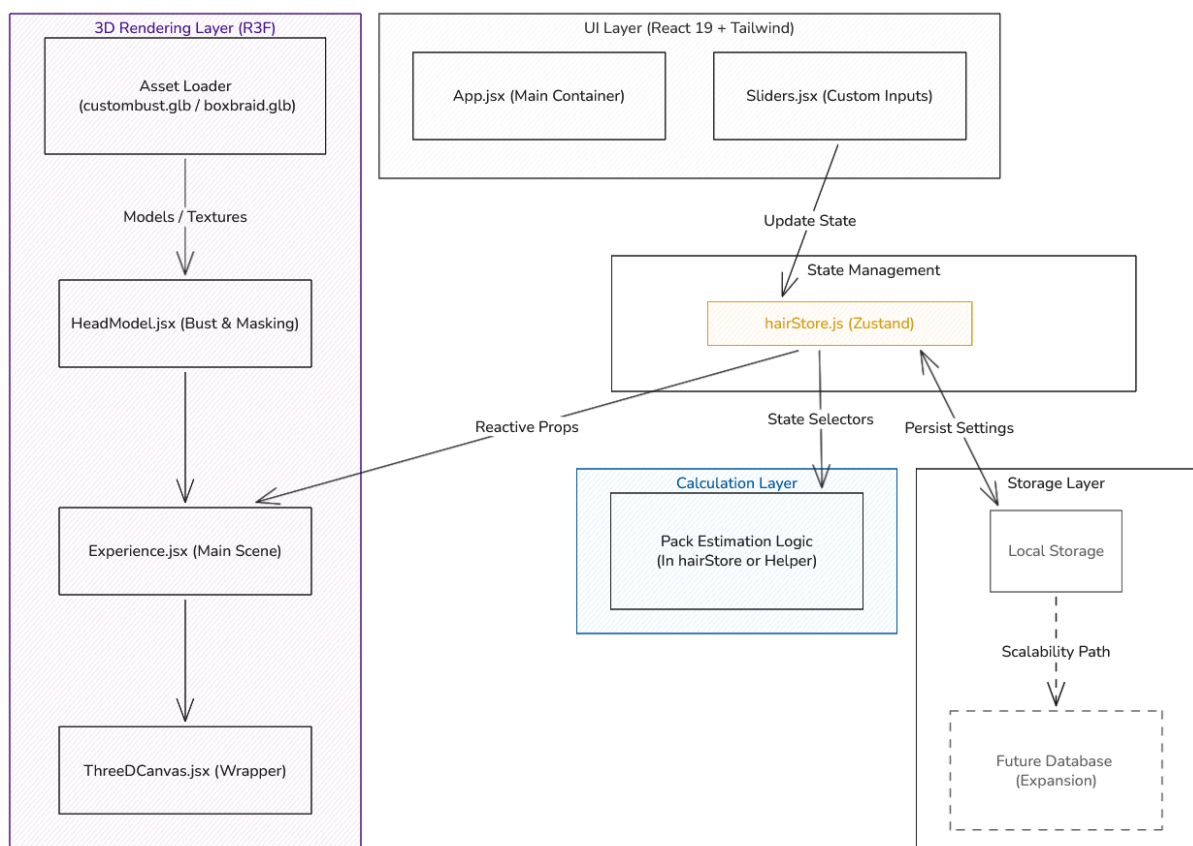
- UI Layer (React Components)
- Zustand State Store
- Calculation Engine
- 3D Rendering Layer (R3F / Three.js)
- Asset Loader (GLTF / Texture Loader)
- Local Storage System
- Future Database Layer (Future Expansion)

Purpose

The diagram will document:

- Data flow between UI and rendering systems.
- State synchronization between sliders and 3D scene updates.
- Separation of calculation logic from rendering logic.
- Scalability path toward future backend/database integration.

UML Component Diagram



3.3 User Flow Diagram

The user flow diagram will visualize how users interact with the application from initial load to hairstyle calculation.

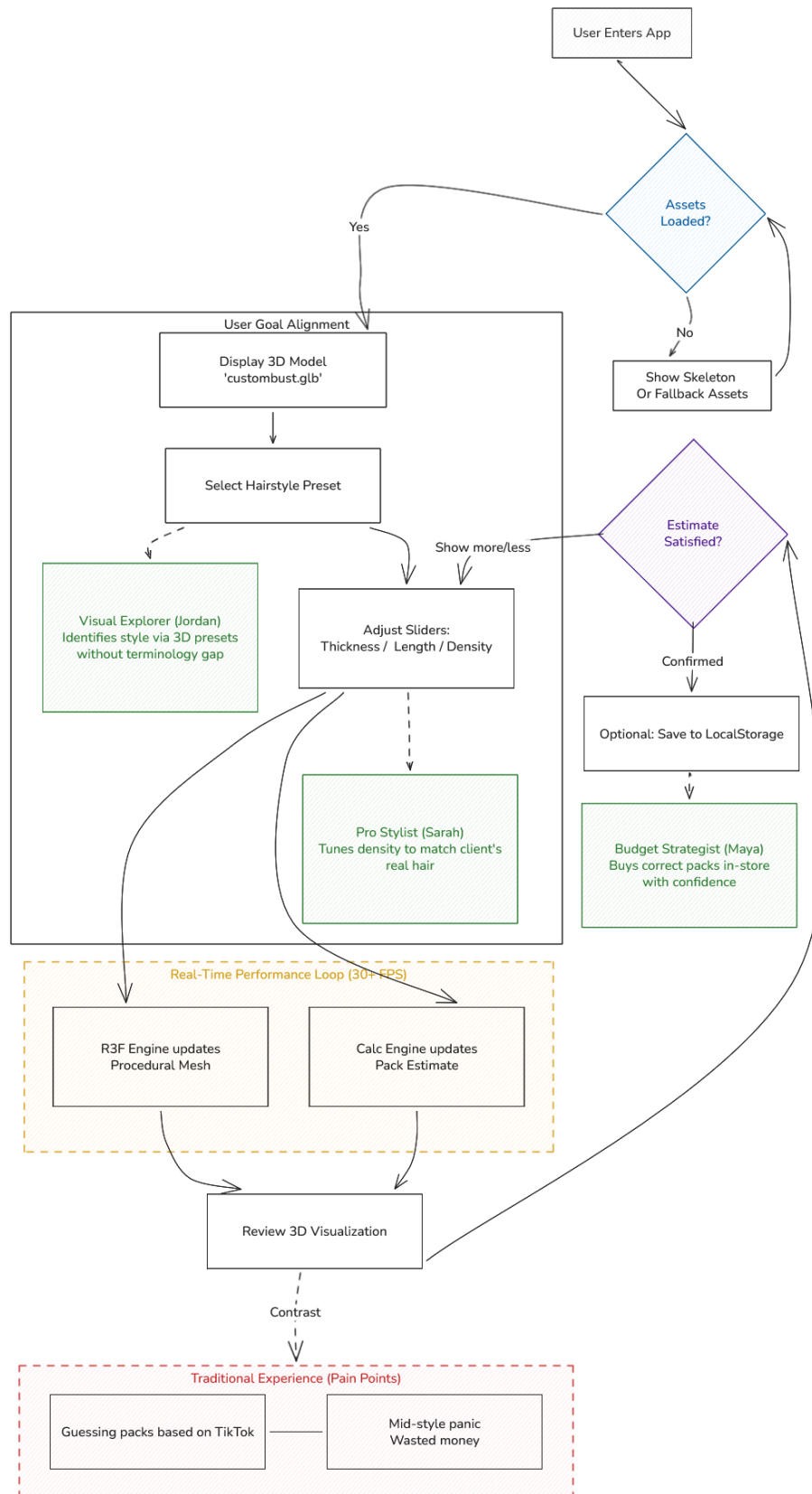
Planned User Flow

1. User opens web application.
2. 3D bust model loads.
3. User selects hairstyle type.
4. User adjusts:
 - Thickness
 - Length
 - Density
5. Procedural hair system updates in real-time.
6. Calculation engine updates estimated pack count.
7. User reviews visualization and estimate.
8. User optionally saves configuration locally.

UX Goal

The application should allow a first-time user to complete the full interaction process in under 2 minutes.

User Flow Diagram



3.4 Rendering Pipeline Overview

The rendering pipeline combines custom 3D asset creation with browser-based procedural rendering.

Pipeline Workflow

1. Hair models and custom bust are modeled in Blender.
2. Models are optimized and exported as **.glb** files.
3. Assets are imported into the React application.
4. React Three Fiber (R3F) manages scene rendering.
5. Three.js/Drei utilities handle:
 - Camera controls
 - Lighting
 - Asset loading
 - Scene helpers
6. Procedural spawning logic places braid instances on the scalp.
7. Zustand synchronizes UI slider state with the 3D scene.

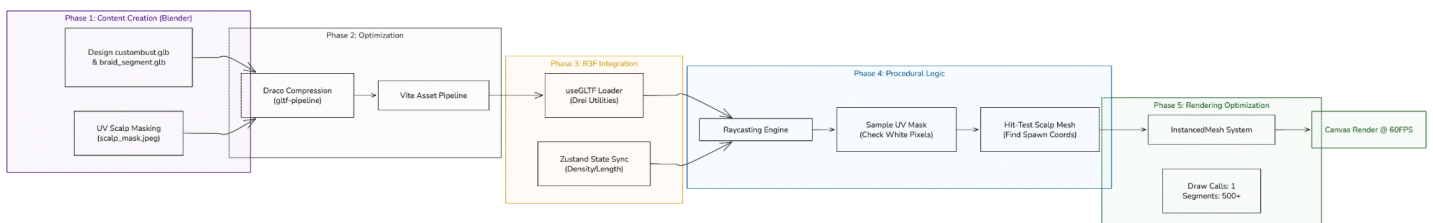
Planned Asset Workflow

Blender > GLB Export > R3F Scene > Three.js Rendering > User Interaction

Optimization Considerations

- InstancedMesh for repeated braid geometry
- GLTF compression using Draco
- Reduced draw calls where possible
- Static braid geometry (no physics simulation)

Rendering Asset Workflow Diagram



3.5 Rendering Strategy Comparison

Two procedural rendering approaches are currently being explored during development.

Comparison Table

Feature	Option 1: Raycasting + UV Masking (Testing Now)	Option 2: Predefined Spawn Points (Considered alternative)
Description	Raycasting projects points onto the scalp; UV maps define "White" (spawn) and "Black" (parting) zones.	Static coordinates are manually or procedurally pre-placed on the scalp mesh.
Advantages	High Dynamism: Procedural placement; realistic parting; highly scalable for new styles.	High Stability: Simple implementation; predictable performance; easy debugging.
Challenges	Higher GPU complexity; harder to debug; potential bottleneck on mobile.	Less realistic; "stiff" parting patterns; requires significant manual setup for each head model.
Mobile Strategy	Lower raycast intervals; dynamic density reduction; InstancedMesh optimization.	Adjust visible instances dynamically based on slider state.
Slider Sync	Sliders live-calculate the Raycast grid density and UV sampling frequency.	Sliders simply toggle the visibility of existing coordinate points.

Development Decision

Both methods will be prototyped during development to determine the best balance between:

- Visual quality
- Real-time performance
- Development scope
- Semester time constraints

3.6 Wireframes / Prototype Space

Wireframes and interface prototypes will be added during Sprint 3 (UI & State Management).

Planned Prototype Screens

- Landing Screen
- 3D Visualization Screen
- Hairstyle Parameter Controls
- Mobile Responsive Layout
- Result / Pack Estimation Screen

Design Goals

- Minimal visual clutter
- Mobile-first usability
- Fast interaction workflow
- Accessible slider controls
- Clear visual hierarchy

Figma prototype:

Figure 1.

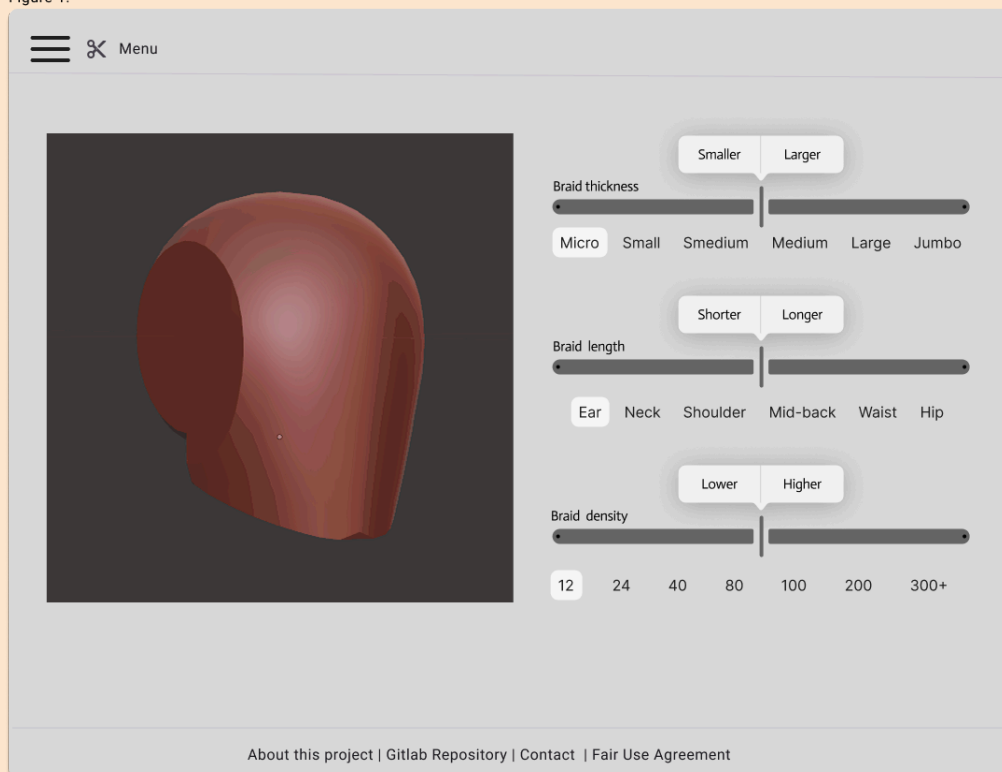
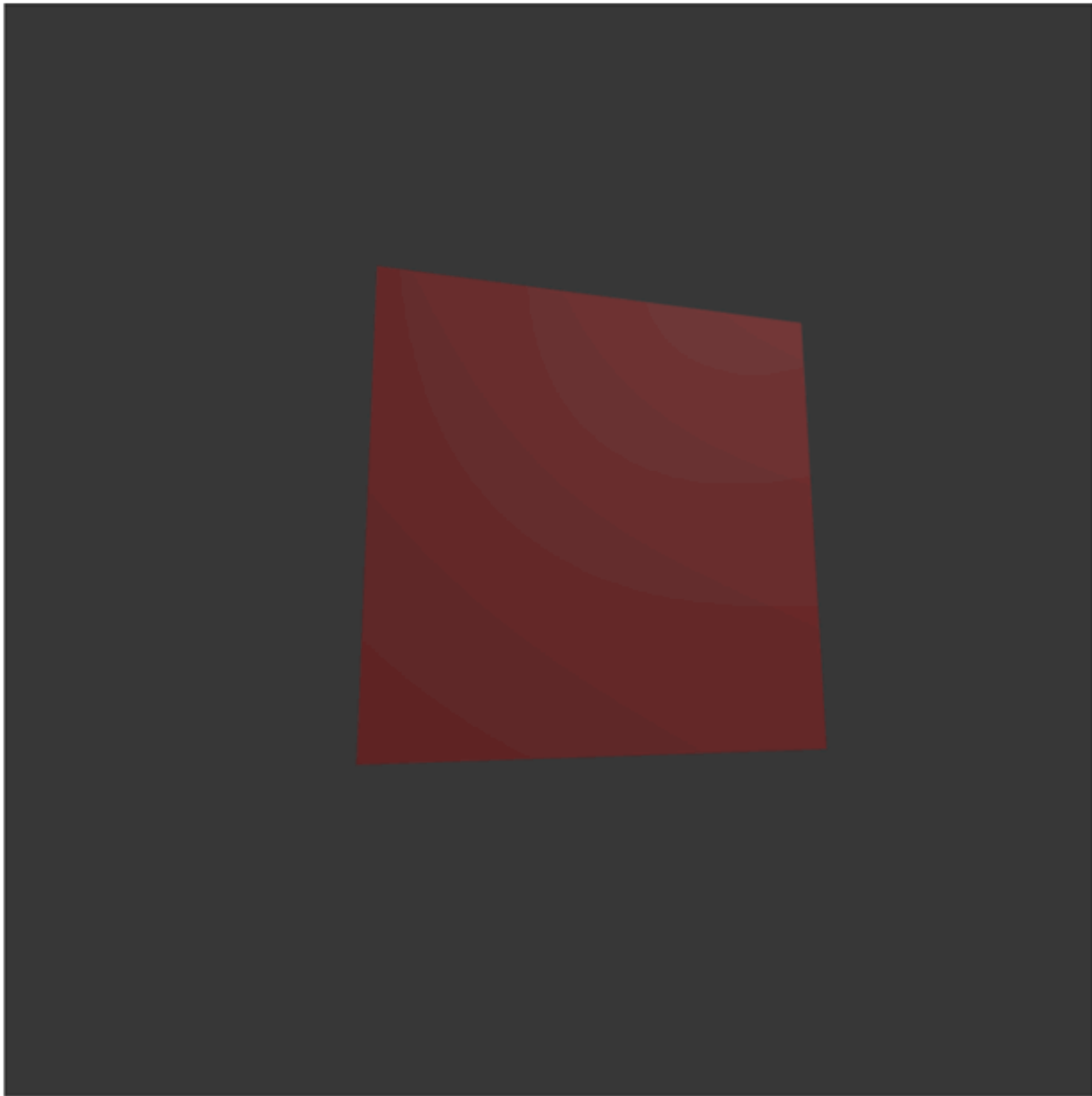


Figure 1.

Test Cube Threejs and basic JavaScript calculation:

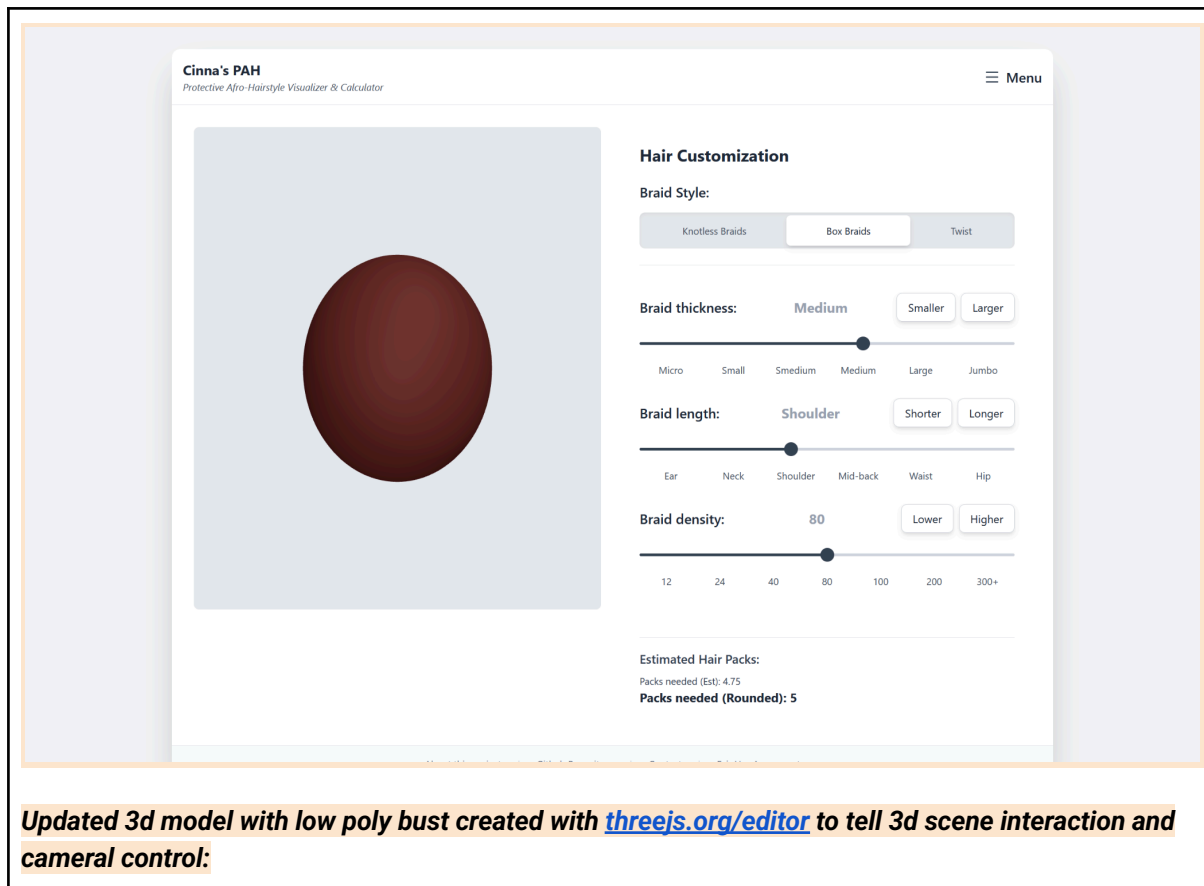


Hair Customization

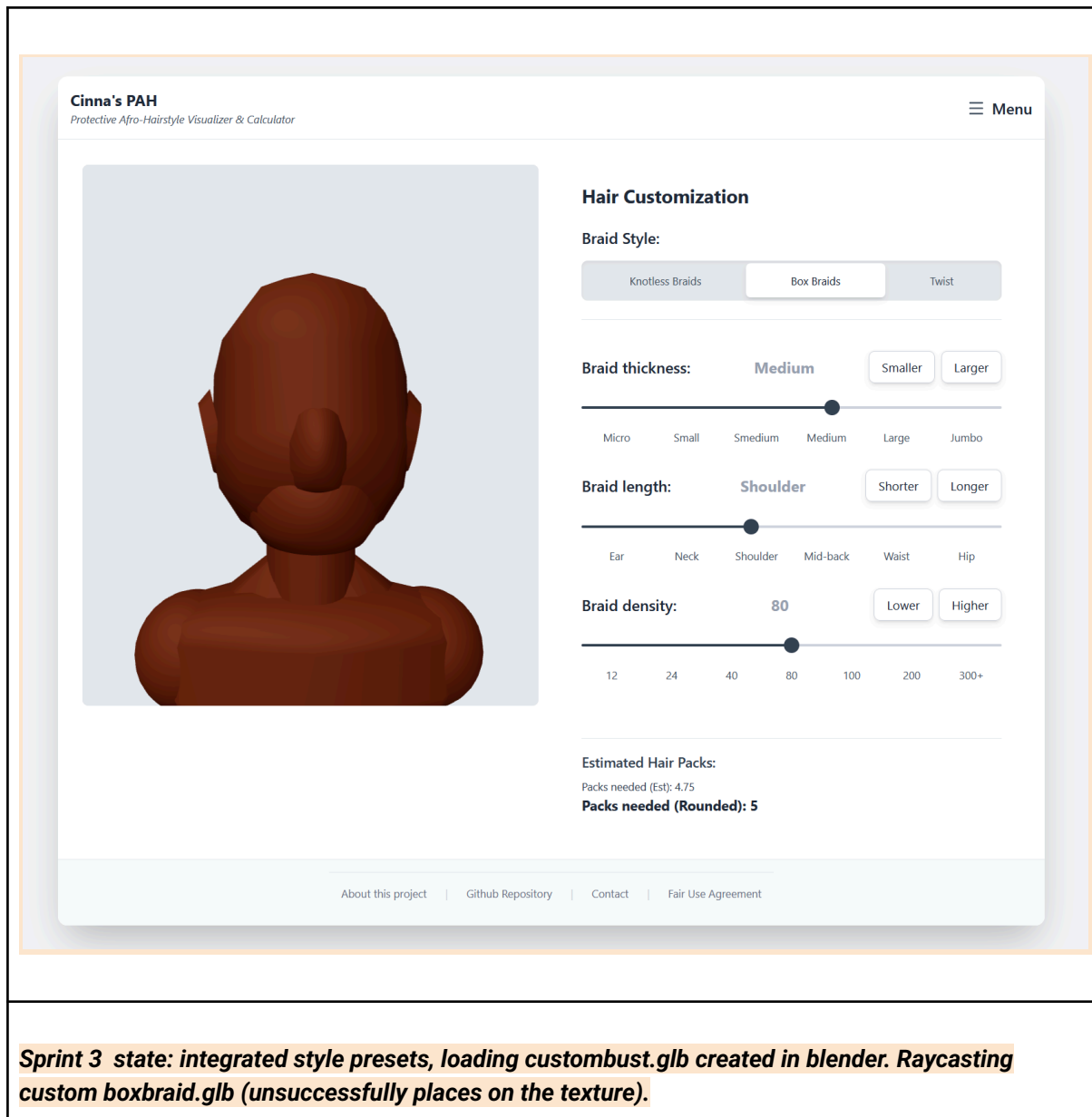
Braid thickness:

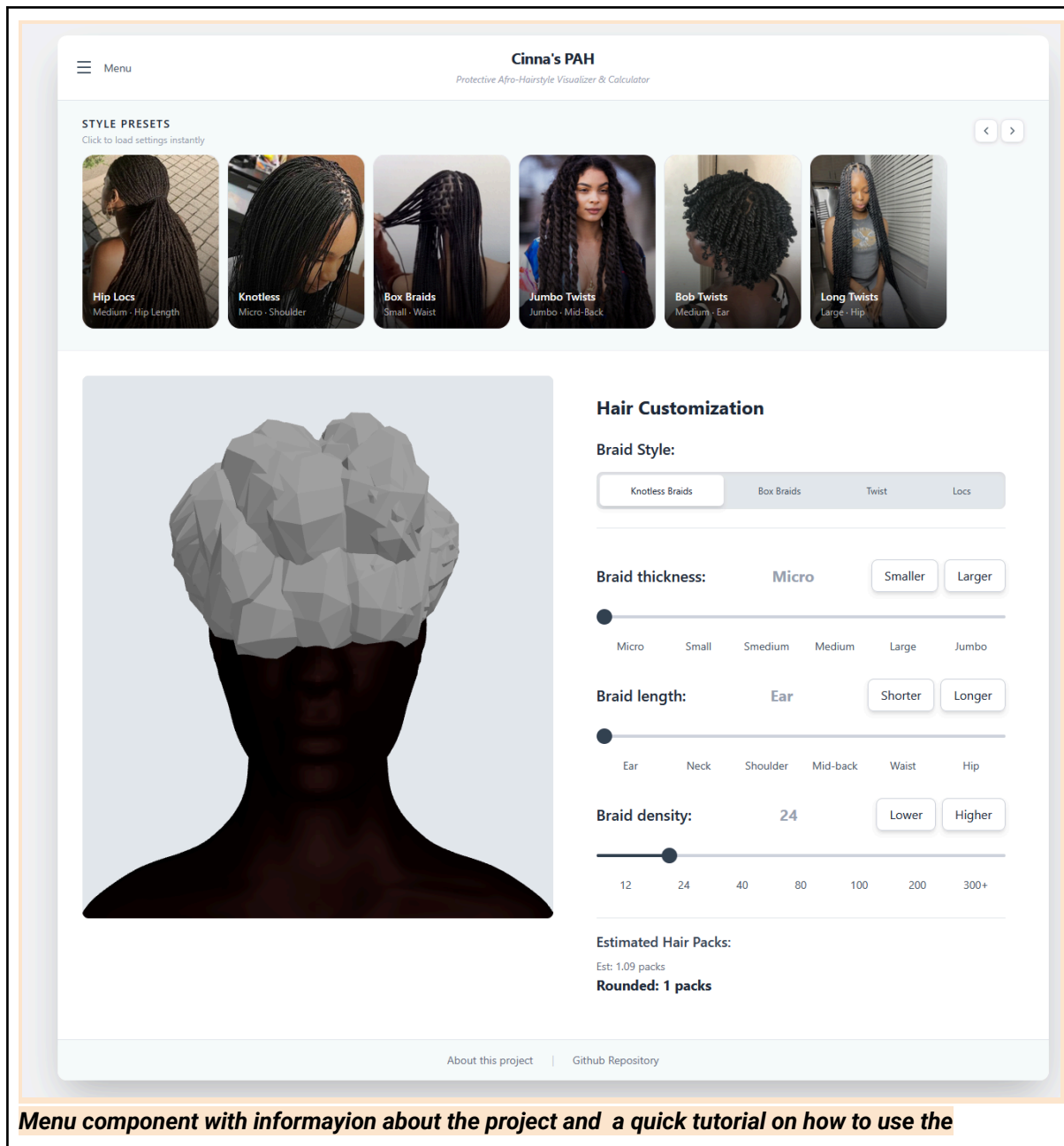
Smaller Larger

Early Implementation using tailwindcss and testing sliders with calculation:



Updated 3d model with low poly bust created with threejs.org/editor to tell 3d scene interaction and cameral control:





Menu component with informayion about the project and a quick tutorial on how to use the

application. As well as an explanation of the calculation behind the pack result and links to GitHub .

Cinna's PAH

Protective Afro-Hairstyle Visualizer

✕

ABOUT THIS PROJECT

Cinna's PAH is a 3D Protective Afro-Hairstyle Visualizer & Calculator with procedural hair placement using raycasting and UV texture masking.

Rotate the 3D head model in real time, dial in your parameters using the sliders, and get an instant pack estimate. Hair strands are procedurally placed based on scalp texture patterns.

HOW IT WORKS

1. Pick a **braid style**

2. Set **thickness** from Micro to Jumbo

3. Choose your desired **length**

4. Adjust **density** (number of braids)

5. Hair placement updates based on scalp texture masking

Calculation formula:

$$(\text{style} \times \text{thickness} \times \text{density}) \times \text{length} \times 0.95$$

Hair placement uses **raycasting** + **UV texture masking**. Rays shot from above the head intersect with the scalp mesh; white pixels in the scalp texture spawn hair, black pixels create parting/skipped areas.

TECH STACK

React 19

R3F

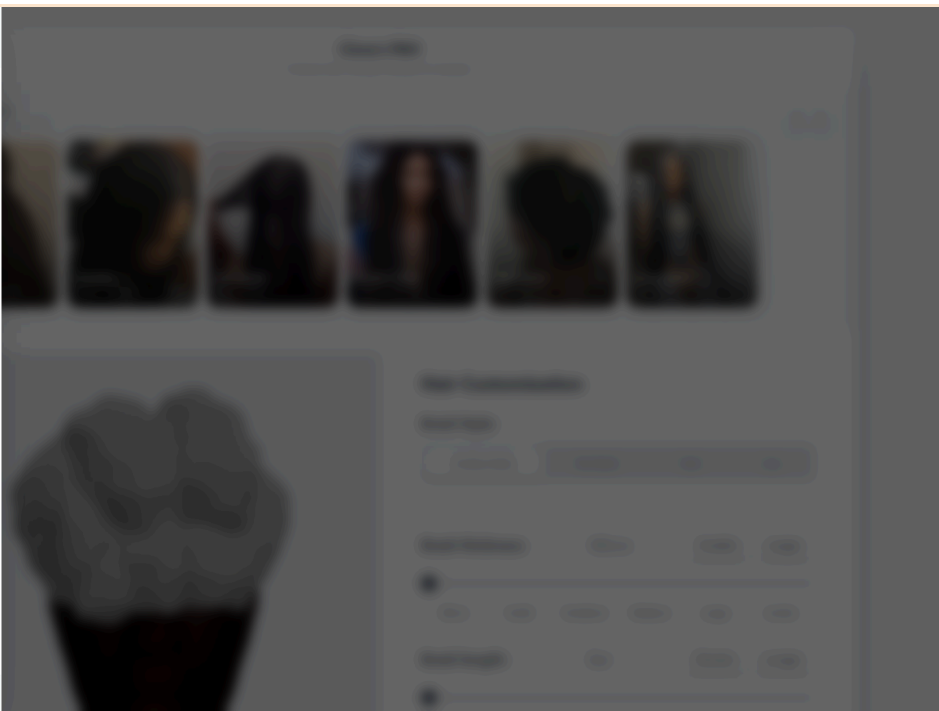
Zustand

Three.js

MIT License

Open Source

GitHub



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3.7 Future Data Model Plan

The current semester implementation will use localStorage and JSON-based configuration data. However, the project is designed with future database integration in mind.

Current Storage

- localStorage
- JSON configuration objects
- Static hairstyle presets

Future Database Goals

Potential future migration:

- Firebase
- Supabase
- PostgreSQL

Planned Data Structure Table

Entity	Field	Type	Description / Logic
User	<i>id</i>	UUID	Primary key for user identification.
	<i>username</i>	String	Display name for the dashboard.
	<i>savedStyles</i>	Array	References to HairstylePreset IDs (CRUD goal).
Hairstyle Preset	<i>id</i>	UUID	Unique identifier for a specific saved configuration.
	<i>styleName</i>	String	"Knotless Braids", "Butterfly Locs".
	<i>density</i>	Number	Mapping to the 1–7 scale used in the UI.
	<i>thickness</i>	Number	Mapping to the 1–7 scale used in the UI.
	<i>length</i>	Number	Modifier used in the calculation.
	<i>estimatedPacks</i>	Number	The final output of the calculation formula.

Validation Feedback	<i>stylistName</i>	String	Name of the professional expert interviewed.
	<i>recommendedPacks</i>	Number	The expert's estimate based on the preset.
	<i>actualPacksUsed</i>	Number	Real-world result from historical data/testing.
	<i>notes</i>	Text	Qualitative insights to tune the calculation logic.

Reason for Exclusion This Semester

A backend/database architecture is intentionally excluded this semester to prioritize:

- Procedural rendering research
- Frontend interaction systems
- 3D performance optimization
- Validation testing

4. Project Organization

Stakeholder

Name	Role	Responsibilities
Osiana "Cinna" Romy	Developer / Designer	Coding, 3D Assets, Planning, UI/UX
Fontys Coach	Stakeholder / Coach	Technical guidance, Academic validation
Professional Stylists	Expert Stakeholders	Calculation accuracy validation

Communication

- **Bi-weekly:** Check-ins with Semester Coach.
- **Ad-hoc:** Feedback loops with the Afro-hair community on Reddit/Discord.
- **Documentation:** Devlog maintained in the project portfolio.

Test Environment and Requirements

- **Browsers:** Chrome, Safari, Firefox (Mobile/Desktop focus).
- **Hardware:** Google Pixel 8 (Performance testing), Windows Asus Zenbook (Development).
- **Tools:** [node.js](#) Dev Server, Three.js Inspector.

Configuration Management

- **GitHub:** Main repository for source code.
- **Branching Strategy:** **main** (stable/deploy), **dev** (feature integration), **feature/3d-logic** (procedural development).
- **Assets:** Versioned in **/public/models** and **/public/textures**.

4.5 Testing Strategy

Testing will focus on lightweight validation appropriate for the semester scope.

Testing Category	Focus Areas	Methodology / Tools
Functional (Alpha)	Ensuring braids spawn strictly on the UV-masked "white" zones. erifying Zustand store updates calculation results instantly.	Visualizing Raycast hit points in the R3F scene using MeshBasicMaterial.
Mobile Performance	Checking if the procedural generation crashes mobile browsers at max density.	Real-world profiling on Google Pixel 8 and remote debugging via Chrome DevTools.
User Acceptance (UAT)	Verifying if users like Jordan (Persona 2) can find "Twists" without knowing the formal term.	Tasking users to "Visualize a short style and find the pack estimate."
Formula Validation	Checking if the 0.95 coefficient holds up against real-world install data.	Inputting data from "Braid Charts" into the engine and comparing outputs.

Planned Workflow

On every push:

1. Install dependencies
2. Run ESLint
3. Run production build
4. Verify deployment readiness

Reason for Limited Automated Testing

The project prioritizes procedural graphics experimentation over enterprise-scale testing infrastructure during this semester.

4.6 Peer Feedback & UX Validation

Peer Feedback Strategy

Stakeholder Group	Feedback Focus	Validation Goal
Classmates (Familiar)	Accuracy of the experience; UX flow.	Does the tool reflect their personal braiding struggles?
Classmates (Unfamiliar)	UI Clarity and technical simplicity.	Is the application technically understandable and visually intuitive for a novice?
Online Community	Feature requests and edge cases.	Does the logic hold up against diverse global braiding styles?
Professional Stylists	Mathematical & procedural accuracy.	Is the 0.95 margin and density scaling clinically accurate for professional use?

Areas of Feedback

Area	Focus
UI Clarity	Ease of understanding controls
Cultural Relevance	Accuracy of hairstyle representation
Accessibility	Readability and interaction flow
Visualization	Trust in hairstyle preview
Calculation Confidence	Trust in pack estimate

4.7 Performance Benchmark Considerations

Performance optimization is an important technical consideration due to the procedural 3D rendering pipeline.

Benchmark Area	Focus Description	Monitoring Tools
FPS Stability	Ensuring a smooth 60 FPS (or stable 30 FPS on low-end devices) during rotation.	Chrome DevTools / R3F Perf
Draw Call Count	Monitoring the efficiency of InstancedMesh to keep draw calls low.	Three.js Inspector
GLTF Loading Time	Optimizing the time it takes to fetch and parse the 3D Bust Asset and hair segments.	Chrome Network Tab
Mobile GPU Stress	Testing thermal throttling and shader performance on mobile browsers.	Chrome DevTools (Remote Debugging)
Spawn Density Scaling	Stress-testing the Raycaster and CPU when the density slider is set to maximum.	R3F Performance Monitor

Why Full Benchmarking is Out of Scope

Comprehensive profiling is not a primary focus this semester because:

1. **Iterative Mindset:** I'm not focusing on delivering a functional interactive system instead of worrying about technical uncertainties.
2. **Condensed Timeline:** Following the pivot in Sprint 3, development time is strictly allocated to core functional requirements to meet Software 2 & UI/UX 1 KPI.
3. **Educational Goal:** The focus is on the learning journey of R3F and procedural generation rather than enterprise-level optimization.

4.8 Comparative Analysis Summary

Comparative research was conducted during the previous semester to evaluate existing hairstyle estimation and inspiration tools.

Existing Solution	User Goal	Common User Frustration	Limitation Compared to This Project
Online Hair Calculators	Quickly estimate how many hair packs are needed before purchasing extensions.	Estimates feel generic and inconsistent between websites. Users cannot visualize the final hairstyle result.	No visualization of style or parting, low personalization, limited control. Results may vary depending on hair packs bought.
Social Media / Forum Feedback (Reddit, TikTok, Discord)	Receive advice from experienced users or stylists who have done similar hairstyles.	Answers are subjective, contradictory, and dependent on personal experience. Users may receive multiple conflicting recommendations.	No standardized calculation system or interactive visualization process. Intrinsic knowledge and terminology between recurring user. Not beginner friendly.
YouTube Tutorials	Follow hairstyle tutorials and estimate materials based on creator demonstrations.	Difficult to translate tutorial results to the user's own head size, density, or preferred braid thickness.	Tutorials are passive learning tools and result may vary depending on the user's own hair type and skills.
Pinterest Inspiration Boards	Collect hairstyle references and communicate aesthetic preferences to stylists.	Users can save inspiration images but still do not know how many packs are required or whether the style is achievable within budget.	Focused on inspiration only, without estimation accuracy.
Hairstyle Catalog / Curation Websites	Browse categorized hairstyle collections and compare style variations.	Categories are often arbitrary or trend-based, making it difficult to estimate practical requirements such as time, density, or material usage.	One image per style, no pack estimation or hair color recommendation.

Comparative Conclusion

At the time of research, no existing platform combined:

- Real-time 3D visualization
- Procedural hairstyle generation
- Interactive density controls
- Hair pack estimation
- Afro-textured hairstyle specialization

This creates a unique opportunity for innovation within the project scope.

5. Finances and Risks

5.1 Cost Budget

- **Vercel Hosting:** Free Tier.
- **GitHub:** Free Tier.
- **3D Assets:** Self-modeled in Blender (0 cost).

5.2 Risks and Mitigation

Risk	Impact	Prevention Activity	Mitigation
Performance Lag	High	Use InstancedMesh for hair segments.	Implement Level of Detail (LOD) or reduce spawn density.
Formula Inaccuracy	Medium	Cross-reference Braid Charts.	Adjust 0.95 factor post-interview.
Technical Debt	High	Pivot from Next.js to pure React early.	Revert to a simplified 2D representation if 3D fails (Back-up Plan).
Mobile GPU limitations	High	Profile on Pixel 8 early.	Limit max instance count on mobile.
GLTF asset size too large	Medium	Use Draco Compression.	Use WebP textures for scalp masks.
Hair overlap artifacts	Medium	Refine UV masking logic.	Implement jitter/randomness offset.
Scope creep	High		Freeze feature set after Sprint 3
Lack of stylist availability	Medium		Use online survey fallback

6. Conclusion

This project combines procedural 3D rendering, UX design, and culturally relevant problem-solving into a practical software solution. The outcome aims to meet KPI for semester in the following categories: Software 2, UI/UX 1 with interactive visualization , and Organizational Process 1 with iterative research-driven development.

Ethical Considerations

This project aims to improve digital representation of Afro-textured hairstyles, which are historically underrepresented in:

- Interactive 3D applications
- Beauty technology tools
- Character customization systems
- Virtual avatar platforms
- Educational visualization software

The project also explores how culturally specific experiences can be translated into accessible digital tools without relying entirely on generalized beauty standards or non-specialized systems.